

your rights as a research subject, you may contact the Chairman of the Committee on the Use of Humans as Experimental Subjects, M.I.T., Room E25-143B, 77 Massachusetts Ave, Cambridge, MA 02139, phone 1-617-253 6787.

I understand the procedures described above. By clicking next, I am acknowledging my questions have been answered to my satisfaction, and I agree to participate in this study.

Interface questions

[Each of these questions has a true or false response which was entered through a radio button. Participants are not able to start the experiment without answering each question correctly.]

Before beginning the experiment, we would like to confirm a few facts through the following comprehension questions. Please answer True or False to the following questions.

- 1) The algorithm's prediction is based on information from both the X-ray scan as well as the clinical history.
- 2) When the algorithm does not show a prediction, it is because the algorithm thinks the pathology is not present.
- 3) The follow up decision refers to any treatment or additional diagnostic procedures that one would conduct based on the findings of the report.
- 4) Two patients with the same probability score for a condition ought to always receive the same "follow-up" recommendation.
- 5) When a condition at a higher level of the hierarchy receives a less than ten percent chance of being present then all the lower level conditions within this branch automatically receive a zero probability of being present.
- 6) If the algorithm says that the probability of a pathology is present with 80% probability, it means that the AI predicts 80 cases out of 100 have the pathology present.
- 7) Suppose your assessment is that the patient definitely has either edema or consolidation, and you believe that edema is twice as likely as consolidation. Then you would assign 66.67% to edema and 33.33% to consolidation:
- 8) I should only indicate pathologies and findings that would be relevant in a radiology report for the patient.

Interface

Figure B.7 is an example of the clinical history indications available to the participating radiologists under the relevant treatment condition. The thoroughness of the information

varies across available information for every patient. Some examples of varying clinical history information are:

1. 68 years of age, Female, chest pain
2. Unknown age, Unknown, Trauma
3. 55 years of age, Male, Order History: Relevant PMH gastroparesis. Presents with vomiting, retching chest discomfort for a duration of today. Concern for PTX, perforated viscus, pneumomediastinum.
4. 74 years of age, Female, s/p unwitnessed fall, r/o rib fx, pna or effusion
5. Trauma
6. 56 years of age, Male, S/P ICD/ Pacemaker insertion / Complete X-ray without lifting arms above shoulders..

Figure B.7: Clinical History Information

Indication

30 years of age, Female, history of hypertension, abnormal EKG, abdominal pain, evaluate for cardiomegaly or mediastinal widening.

Vitals

Variable	Value
Weight	170 lbs
BP	243/166 mmHg
Temp	99.1F
Pulse	99.0 bpm
Age	30

Abnormal Labs

All Labs

Variable	Value	Unit	Flag
ALT (SGPT), Ser/Plas	38.0	U/L	High
AST (SGOT), Ser/Plas	39.0	U/L	High
Eosinophil, Absolute	0.01	K/uL	Low

Note: The clinical history information environment in the experiment had information on patient indications, vitals and abnormal labs.

Figure B.8: Interface Slider

Airspace OpacityAI Prediction:  **12% (Very unlikely)**

Probability of Airspace Opacity: 43%

Size

☐ Small☒ Medium☐ Large☐ Very Large

Recommend follow up

☒ Yes☐ No

Note: The participants use the slider to indicate the probability of a pathology being present for a given patient based on the treatment offered. For prevalence greater than 10% the participants are required to indicate the prevalence of a sub-pathology (if it exists) and whether a follow-up is recommended.